

أسئلة سنوات سابقة

المادة: الزخافات -

Q1:- Consider the random process $x(t) = A \cos(\omega t)$ when ω is constant and A is a random variable uniformly distributed on $[-1, 1]$.

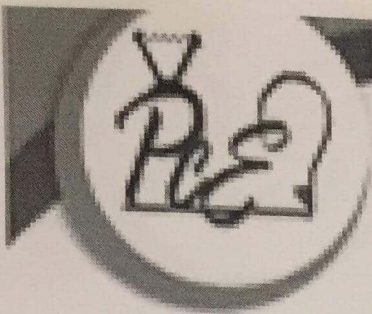
[A] Find mean value and variance For the random process.

[B] Find auto correlation Function of $x(t)$ and determined whether the process is wide sense stationary or not.

[C] determine whether the random processes $x(t)$ and $y(t) = A \sim U(-1, 1)$ are orthogonal or not (Justify).

Q2:- For a random process with autocorrelation Function $R_{xx}(\tau) = e^{-|\tau|}$, Find the mean and the variance of the process (knowing that the process is stationary ergodic with no periodic components).

Q3:- suppose that the monthly income in a big city is normally distributed with mean 500 JD and standard deviation 2, Find the probability that a selected person at random would earn less than 502 JD per Month.



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المادة:

Q4:- let $X \sim \text{Poisson}(m)$ For some $m > 0$
if $P(X=1) = P(X=2)$ Find $P(X=3)$.

Q5: A random variable X with pdf $f(x) = 1$
 $0 < x < 1$ zero else when Find the pdf of
the transformation $Y = e^X + 1$.

Q6: A random variable X with $E(X) = a$ and $E(X^2) = 2a^2$
when $a > 0$ show that $Y = \frac{1}{2}X - a$ are
orthogonal, determine whether X and Y correlated or
not.

Q7:- Two random variable X with pdf $f_X(x) = u(x)e^{-x}$ and
 $Y \sim U(0,1)$ Find the pdf of $W = X + Y$.

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المادة:

Q1:

$$[A] \quad E(x(t)) = \int_{-\infty}^{\infty} x(t) f(A) dA$$

$$f(A) = \begin{cases} \frac{1}{2}, & -1 < A < 1 \\ 0, & \text{o.w} \end{cases}$$

$$= \int_{-1}^1 A \cos \omega t \cdot \frac{1}{2} \cdot dA$$

$$= \cos \omega t \cdot \frac{A^2}{4} \Big|_{-1}^1 = \cos \omega t \left(\frac{1}{4} - \frac{1}{4} \right) = 0.$$

Constant.

$$\text{Var} X = E X^2 - (E X)^2$$

$$E(x(t)^2) = \int_{-\infty}^{\infty} x^2(t) f(A) dA = \int_{-1}^1 A^2 \cos^2(\omega t) \frac{1}{2} dA = \frac{A^3}{6} \cos^2(\omega t) \Big|_{-1}^1$$

$$= \cos^2(\omega t) \left(\frac{1}{6} + \frac{1}{6} \right)$$

$$= \frac{1}{3} \cos^2(\omega t)$$

$$\text{Var}(X) = \frac{1}{3} \cos^2 \omega(t) - 0$$

$$= \frac{1}{3} \cos^2(\omega t)$$

[B] $E(x(t)) = 0 \rightarrow \text{Constant} \rightarrow \text{First order stationary}.$

$$R_{xx}(t, t+\tau) = E(x(t) \cdot x(t+\tau))$$

$$= E(A \cos \omega t) \cdot A \cos(\omega t + \omega \tau)$$

$$= E\left(\frac{A^2}{2} (\cos(2\omega t + \omega \tau) + \cos(\omega \tau))\right)$$

$$= E\left(\frac{A^2}{2} \cos(2\omega t + \omega \tau) + \cos \omega \tau\right)$$

$$= E\left(\frac{A^2}{2} \cos 2\omega t + \omega \tau\right) + E\left(\frac{A^2}{2} \cos \omega \tau\right)$$

$$= \int_{-1}^1 \frac{A^2}{2} \cos(2\omega t + \omega \tau) \frac{1}{2} dA + \int_{-1}^1 \frac{A^2}{2} \cos(\omega \tau) \frac{1}{2} dA$$

$$= \frac{A^3}{6} \cos(2\omega t + \omega \tau) \frac{1}{2} \cdot dA + \int_{-1}^1 \frac{A^2}{2} \cos(\omega \tau) \frac{1}{2} dA.$$

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المادة:

[B] →

$$= \frac{1}{6} \cos(2\omega t + \omega \tau) + \frac{1}{6} \cos(\omega \tau) \neq R_{xx}(\tau)$$

So, isn't 2nd order stationary.

$x(t)$ isn't wide-sense stationary.

[C]

$x(t)$ is orthogonal if $E(xy) = 0$

$$E(A \cos(\omega t) \cdot A)$$

$$E(A^2 \cos(\omega t))$$

$$\int_{-1}^1 A^2 \cos(\omega t) \cdot \frac{1}{2} \cdot dA = \frac{A^3}{3} \cos(\omega t) \frac{1}{2} \Big|_{-1}^1$$

$$= \frac{1}{3} \cos(\omega t) \neq 0 \quad \text{so, isn't orthogonal.}$$

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المادة:

Q2:-

$$R_{XX}(\tau) = e^{-|\tau|}$$

$$EX = \bar{X}$$

$$R_{XX}(0) = e^{-0} = 1 = EX^2$$

$$\lim_{\tau \rightarrow \infty} R_{XX}(\tau) = e^{-\infty} = \text{zero} = \bar{X}^2$$

$$EX = \text{zero}$$

$$\text{Var} = EX^2 - (EX)^2 = 1 - 0 = 1$$

Q3:-

$$Z = \frac{X - M}{\sigma} = \frac{502 - 500}{2} = \frac{2}{2} = 1$$

$$T(1) = \boxed{}$$

من الجدول

Q4:-

$$P(X=1) = P(X=2)$$

$$\frac{e^{-m} m^1}{1!} = \frac{e^{-m} m^2}{2!}$$

$$m = \frac{m^2}{2}$$

$$m^2 - 2m = 0$$

$$m(m-2) = 0$$

$$m = 0 \quad \boxed{m = 2}$$

$m > 0$
من السؤال

$$\begin{aligned} P(X=3) &= \frac{e^{-m} m^3}{3!} \\ &= \frac{e^{-2} 2^3}{3!} \\ &= \boxed{0.180} \end{aligned}$$

أسئلة سنوات سابقة

المادة:

Q5:-

$$f(x) = 1$$

$$y = e^x + 1$$

$$e^x = y - 1$$

$$x = \ln y - 1$$

$$f_y(y) = f_x(\ln y - 1) \frac{d(\ln y)}{dy}$$

$$= (1) \frac{1}{y-1}$$

$$f_y(y) = \frac{1}{y-1}$$

$$\frac{dx}{dy} = \frac{1}{y-1}$$

Q6:-

$$E_{xy} \stackrel{?}{=} 0$$

$$\begin{aligned} E_{xy} &= E\left(\frac{1}{2}x - a\right) \\ &= E\left(\frac{1}{2}x^2 - xa\right) \\ &= \frac{1}{2} E x^2 - a E x \\ &= \frac{1}{2} (2a^2) - a a \\ &= a^2 - a^2 = \text{zero} \checkmark \end{aligned}$$

yes, orthogonal.

$$E_x = a$$

$$E_y = E\left(\frac{1}{2}x - a\right)$$

$$= \frac{1}{2} E x - a$$

$$= \frac{1}{2} a - a = -\frac{1}{2} a$$

↓ $E_{xy} \stackrel{?}{=} E_x E_y$

$$0 \neq a \left[-\frac{1}{2}a\right]$$

$$0 \neq -\frac{1}{2}a^2$$

yes, correlated.

أسئلة سنوات سابقة

المادة:

Q7:

$$f_w(w) = \int_{-\infty}^{\infty} f_y(y) f_x(w-y) dy$$

$$= \int_{-\infty}^{\infty} 1 \cdot u(w-y) e^{-(w-y)} dy$$

$$= \int_0^w e^{-(w-y)} dy = \frac{1}{1} e^{-w+y} \Big|_0^w$$

$$= e^0 - e^{-w}$$

$$= 1 - e^{-w}$$

Q1:- If $f_x(x) = \frac{1}{a} [u(x) - u(x-a)]$
 and $f_y(y) = b u(y) e^{-by}$, $(a < b)$, $(a, b > 0)$
 find The density of $w = x + y$ s.t x, y are ind.?

soll:-

$$f_w(w) = \int_{-\infty}^{\infty} b u(y) e^{-by} \frac{1}{a} [u(w-y) - u(x-y-a)] dy$$

$$= \frac{b}{a} \int_0^{\infty} e^{-by} [u(w-y) - u(w-y-a)] dy$$

$$= \frac{b}{a} \left[\int_0^{\infty} e^{-by} u(w-y) dy - \int_0^{\infty} e^{-by} u(w-a-y) dy \right]$$

$$= \frac{b}{a} \left[\int_0^w e^{-by} dy - \int_0^{w-a} e^{-by} dy \right]$$

$$= \frac{b}{a} \left[-\frac{1}{b} e^{-by} + \frac{1}{b} e^{-b(w-y)} \right]$$

$$f(w) = \frac{1}{a} e^{-b(w-y)} - \frac{1}{a} e^{-by}$$

$$f(w) = \begin{cases} \frac{1}{a} e^{-b(w-y)}, & 0 < w < a \\ \frac{1}{a} e^{-b(w-y)} - \frac{1}{a} e^{-by}, & w \geq a \end{cases}$$

Q2) Let X be a r.v such that :-

$$E X = 3, \quad G^2 X = 2 \quad \text{and} \quad Y = -6X + 22$$

find :- $E Y, R_{xy}, \text{Cov}(X, Y)$

and show if X, Y are indep. ??

Sol :-

$$Y = -6X + 22$$

$$E Y = -6 E X + 22 = -6(3) + 22 = 4$$

$$E Y = 4$$

$$E X Y = E (X (-6X + 22))$$

$$= E (-6X^2 + 22X)$$

$$E X Y = -6 E X^2 + 22 E X$$

$$\text{but } G X^2 = \text{Var } X = E X^2 - (E X)^2$$

$$2 = E X^2 - (3)^2$$

$$E X^2 = 11$$

$$E X Y = -6(11) + 22(3) = 0$$

so, X, Y are orthogonal

$E X Y \neq (E X)(E Y) \neq (3)(4) \neq 0$ so are not independent

$$\text{Cov}(X, Y) = -(E X)(E Y) = -(3)(4) = -12$$

Q3) Let $X(t)$ be a wide-sense stationary with $R_{XX}(\tau) = e^{-\alpha|\tau|}$ (auto correlation). Assume that $X(t) = \cos(\omega_0 t + \theta)$ where θ is uniform over $(-\pi, \pi)$ which is indep. of $X(t)$. Let $Y(t) = X(t) \cos(\omega_0 t + \theta)$, find the auto-correlation func. of $Y(t)$?

Sol:-

$$\begin{aligned}
 R_{YY}(t, t+\tau) &= E[Y(t)Y(t+\tau)] \\
 &= E[X(t)X(t+\tau)\cos(\omega_0 t + \theta)\cos(\omega_0 t + \omega_0 \tau + \theta)] \\
 &= R_{XX}(t, t+\tau) \frac{1}{2} E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta) + \cos(\omega_0 \tau)] \\
 &= \frac{1}{2} R_{XX}(\tau) [E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] + E[\cos(\omega_0 \tau)]] \\
 &= E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] = \int_{-\pi}^{\pi} \frac{1}{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta \\
 &= \frac{1}{2} e^{-\alpha|\tau|} \cos(\omega_0 \tau) = R_{YY}(\tau)
 \end{aligned}$$

So, $Y(t)$ is wide-sense stationary

Q4) Let The two random Processes

$$X(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

$$Y(t) = B \cos(\omega_0 t) - A \sin(\omega_0 t)$$

are k.v's, (ω_0) is constant A, B are uncorrelated, zero-mean with same variance, find the cross correlation $R_{xy}(t, t+\tau)$ and show that $X(t), Y(t)$ are jointly wide-sense?

Sol:-

$$* \text{ Zero mean } = E A = E B = 0$$

$$* \text{ Same var } = \sigma_A^2 = \sigma_B^2$$

$$\Rightarrow R_{xy}(t, t+\tau) = E(X(t)Y(t+\tau))$$

$$= E((A \cos \omega_0 t + B \sin \omega_0 t)(B \cos(\omega_0 t + \omega_0 \tau) - A \sin(\omega_0 t + \omega_0 \tau)))$$

$$= E(A B \overset{1}{\cos \omega_0 t} \overset{2}{\cos(\omega_0 t + \omega_0 \tau)} - A^2 \overset{3}{\cos \omega_0 t} \overset{4}{\sin(\omega_0 t + \omega_0 \tau)} + B^2 \overset{3}{\sin \omega_0 t} \overset{4}{\sin(\omega_0 t + \omega_0 \tau)} - A B \overset{1}{\sin \omega_0 t} \overset{2}{\sin(\omega_0 t + \omega_0 \tau)})$$

$$= E(B^2 \sin(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau) - E(A^2 \cos(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)))$$

إنتهى

Q4)

$$E(AB) = E(A)E(B) = 0 \Rightarrow \text{zero mean}$$

$$\sigma_A^2 = \sigma_B^2 = \sigma^2$$

$$E A^2 - (E A)^2 = E B^2 + (E B)^2 = \sigma^2$$

$$E A^2 = E B^2 = \sigma^2$$

Then $R_{xy} = -\sigma^2 \sin(\omega_0 \tau)$ بعد التعويض بالخطية

so $y(t), x(t)$ are jointly - sense stationary

* بالامانة لسؤال تكامل بسيط

* سؤال اختيار متعدد (10 دوائر) اسئلة مكررة من الامم

وباقى الاسئلة على مواقع سفلة اخرى مثل

Binomial, Poisson, normal distribution
standard normal,

التوفيق

$$① \quad P_{xy}(x,y) = \begin{cases} xy/36 & \text{for } 0 < x < 4, 0 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find

- ① The marginal density function $P_x(x)$
- ② The conditional density function $P_y(y/x)$.
- ③ The ~~joint~~ distribution function $P_{xy}(x,y)$.

5/9/20

$$② \quad P_x(x) = 5e^{-5x} \quad x \geq 0$$

① Find $\phi(\omega)$ of this function.

② random variable x

③ average value of R.V. x .

5/9/20

given if $P(A) = P(\bar{A})$ we say ?

I guess $A = \phi$

6/9/20

a random variable is random number.

I guess always true

7/9/20

if $C_{xy} = 0$ then x and y what?

$$\text{if } R_{xx}(\tau) = 16 + \frac{16(1 + \sin(2\pi\tau))}{16 + \tau^2} \quad \text{assume wide sense stationary}$$

↳ is fun periodic or non —

↳ find $\bar{x}$

↳ find variance of random $x(t)$.

السؤال الرابع -

سؤال مباشر (7) ب 22 علامة

$$x(t) = A \sin \omega t$$

$$y(t) = 4 \cos(100t + \phi)$$

1) Find $E x(t)$ and R_{xx}

11

2) Find $E y(t)$ and R_{yy}

3) Prove if $x(t)$ are WSS or not

4) Prove if $y(t)$ are WSS or not

5) Prove if $x(t)$ ergodic or not

6) Prove if $y(t)$ ergodic or not

7) Find C_{xy}

uniformly distributed (y(t)) - مسألة
distribution

التي $x(t)$ لها L_p اقتران

$$P_a = \frac{k}{A}$$

1. T

2. Dependent on the case 3. F

Under full event A & B if they are given mutual exclusive
this world given that $P(A), P(B) \neq 0$

1. Not Connection idempotence & Mutual exclusive
2. Not Statically independent
3. Statically independent
4. No enough info to determine

YES
PES

All of the given are considered as a probability of - conditional
except:-

قابل التامة

- السؤال الاول

لا دوائر

1) when $C_{xy} = 0$

2) $P(A) + P(B) \leq$

3) if $P(A) = P(\bar{A})$ then: A) $A = \emptyset$ B) $A \cap S$

C) $A = S$

4) Properties of joint distribution function

سوالين True and false

- السؤال الثاني

$$R_{xx} = \frac{25 + \frac{4}{1+6x^2}}{1+6x^2}$$

بعض اقتران

1. Find $|X|$

بعض دوائر

2. Find the power

4. Find ϕ^2

5. Function periodic or not

- السؤال الثالث



مقدار $P(k_1), P(k_2), P(g_1/k_1), P(g_2/k_1)$

1) $P(k_1/g_1)$ 2) $P(k_2/g_1)$

المطلوب

3) $P(g_1)$ 4) $P(g_2)$

In S-T Sample Space if the probability of event is $= 0$
this event may!!



1. No event has prob. $= 0$
2. Occurs even if the prob. $= 0$
3. No enough information to determine
4. May never occur

under Certain Condition it is possible that the sum of the probability of points in Sample Space S is less than one

1. T

9. Depend on the data 3. F

In wireless system, if the received power level is assumed to follow the following pdf with random variable X in (mW) as given below, find

1. The probability density $f_X(x)$
2. The value of a for $f_X(x)$ to be valid p.d.f.
3. Compute P_S the prob. the received power signal (P_{signal}) is at least 10 mW

$$f_X(x) = \begin{cases} 1 - (ae^{-x/5}) & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Ans

3. Statically independent.
4. No enough data to disprove.

All of the given are considered as a property of cdf, except:

Ans

1. $F_X(100) \leq 1$
2. $0 \leq F_X \leq 1$
3. $F_X(x_1) \times F_X(x_2) \leq x_1 \times x_2$
4. $F_X(x-1) \leq F_X(x)$